M0 Project Proposal

Due Fri Oct 30, 11:59 PM ET · One submission per group on Gradescope

Type your answers in the boxes. Boxes do not grow, and text that does not fit will be cut off. Submit both pages and do not add, remove or reorder pages. Before uploading, use Print to PDF (or save a flattened copy) so your answers stay visible on Gradescope.

1 Team

Names and Andrew IDs of all three members.

Member 1 name	Andrew ID	Member 2 name	Andrew ID	Member 3 name	Andrew ID
Ayush Jain	ayushj3	Bhuvana Kanakam	bkanakam	Prithvi Sunilkumar	psunilku

2 Project name and summary

In two or three sentences, what does your system do, and who uses it?

Project name

Giraffe: One Board for Every Workflow

Summary

Giraffe lets groups create shared, customizable boards to organize work, tasks, and opportunities. The same board engine supports many uses, such as software sprints, referral openings, and shared household chores. Members set up their own stages, add items, claim or assign owners, and track progress together, while the server keeps the board consistent even when many people update it at once.

3 Shared state

What data does the server hold that multiple clients read and change? Be specific (for example, 'the number of remaining seats for each show, and the list of bookings').

The server stores each board's name, workflow stages, members, and items. Each item has a type, title, description, stage, assigned owner, and activity history. Boards can also store workflow-specific limits, such as the number of referral slots on an opening or a chore's due date.

4 Client operations

List at least three operations a client can perform. At least two of them must change the shared state; mark Yes for each one that does.

	Operation (name and one line on what it does)	Changes state? (Yes/No)
1.	Create a board with a name and workflow stages	Yes
2.	Add, edit, or delete an item, including moving it between workflow stages	Yes
3.	Assign or claim an item's owner	Yes
4.	Add a comment to an item's activity history	Yes
5.	View a board and its items	No

5 The conflict

Describe one concrete scenario in which two clients acting at the same moment could produce a wrong result if the server is not careful. Walk through it step by step.

An unassigned chore, "Take out trash," is on a shared roommate board.

1. Alice clicks "claim." The server checks the item and sees no owner.
2. At the same moment, Bob clicks "claim." The server checks the same item and also sees no owner.
3. Alice's claim is saved, setting the owner to Alice.
4. Bob's claim is saved, overwriting the owner to Bob.

Both were told their claim succeeded, but only one is recorded, so Alice believes the chore is hers while the board says it is Bob's. The same race on a referral opening with one slot left lets two applicants both be accepted.

6 The invariant

State one rule about your shared state that must always be true, no matter how many clients act at once (for example, 'a show never has more bookings than seats'). Your M3 demo will prove this rule holds.

Each item has at most one owner at a time, and no board limit is ever exceeded (for example, a referral opening never accepts more requests than its slot count).

7 Main classes

Name the three to five classes you expect to write, with one line each on what they represent. This is a starting plan to check your thinking, not a commitment: your final classes can change as you build the project.

	Class name	What it represents
1.	Board	owns a board's stages, members, and items; every change goes through its methods, which enforce the invariant
2.	BoardItem	a unit of work with its own validation rules, with subclasses Task, ReferralRequest, and Chore (abstract base class)
3.	WorkflowStage	a named stage that items move through (dataclass)
4.	User	someone who can access and update boards (dataclass)
5.	BoardServer	receives client requests and applies them to the shared boards safely (added in M2)

8 Stretch idea (optional)

One feature you would add if you had extra time.

Reusable templates for common workflows (sprint, referrals, chores) that users can still customize with their own stages and fields.